Cosmos synthetic-data adaptation

Implementation, pilot training, and measured effects

Technical report | VyFlare | 30 October 2026 (America/Chicago)

We added still-image supervision to the single-Spark LoRA runner, prepared a deduplicated synthetic-image dataset with verified real-video replay, and completed a separate 20-update continuation of the Version 1 adapter. The original base, source adapter, and browser tester were preserved. [E1-E4]

Development measure	Base	Version 1, before	New adapter
Synthetic render-control exact class	8/12 (66.7%)	0/12 (0.0%)	9/12 (75.0%)
Real-video exact color sequence	0/9 (0.0%)	4/9 (44.4%)	4/9 (44.4%)
Real-video exact final color	3/9 (33.3%)	5/9 (55.6%)	4/9 (44.4%)
Synthetic schema validity	12/12 (100.0%)	0/12 (0.0%)	12/12 (100.0%)

Secondary exact class-name agreement when an exact bare-word answer is accepted despite missing JSON: base 8/12, Version 1 8/12, continued adapter 9/12. This separates naming from formatting; it is still agreement with renderer controls, not factory accuracy.

On this diagnostic subset, synthetic control-name matches changed from 0/12 to 9/12, while real exact sequences changed from 4/9 to 4/9. These counts describe this run only. The synthetic scene remains fixed and the real benchmark is the previously inspected same-session development partition.

Release decision: retain Version 1 in the tester. Real final-color matches fell from 5/9 to 4/9. The synthetic class-name gain ignoring formatting is one additional case (8/12 to 9/12), so this run does not justify a claim of better factory performance or automatic promotion.

Meaning of the result

Synthetic render-color prediction is a separate task from observed real color sequences. Neither measures chemical identity, mixture ratio, contamination, calibrated volume, recipe correctness, or production benefit. This mixed replay pilot demonstrates the training path and measures a before/after effect; it is not a matched real-only versus mixed-data causal experiment.

Deliverables

Updated image/video runner; source-hashed synthetic preparation; split, target, and metric contracts; automated regression checks; a new saved adapter with logs and before/after outputs; reproduction instructions; this report. The new adapter has not been promoted to the browser service.

1. What changed and why

Change	Implementation	Purpose
Image input branch	Pillow EXIF transpose and RGB conversion; native image tokens, pixels and image_grid_thw. No repeated-frame video conversion.	Train directly on supplied PNGs with the existing Cosmos visual encoder.
Separate target contracts	Synthetic render_color_class string from 12 classes; real observed_color_sequence list. Separate metrics by target kind.	Avoid claiming a renderer parameter is a chemical or a human-verified physical observation.
Continue adapter weights	--initial-adapter loads PEFT weights as trainable; checks source base snapshot and records source adapter hash. Fresh optimizer.	Extend the existing adapter while retaining its source file for rollback. This is not full optimizer-state resume.
Modality-balanced replay	--balance-modalities alternates image and video microbatches.	Keep the much larger synthetic corpus from dominating all presentations; measure possible loss of old behavior.
Independent image budget	--image-max-pixels defaults to 262,144, separate from total-video pixel budget.	Bound image memory without changing historical video preprocessing.
Data and loss checks	Hash/path checks, explicit split groups, cross-partition duplicate rejection, reviewed real targets, image/video preflight, assistant-only loss boundary.	Fail early on changed inputs, leakage, unsupported labels, or incorrect supervision.

Unchanged components

The base reasoner and visual encoder remain frozen. Only language-attention q/k/v/o LoRA parameters train. Rank 8, alpha 16, dropout 0.05, BF16 base, SDPA, and gradient checkpointing remain as in Version 1. The original video-only invocation remains supported. No fluid solver, defect detector, ingredient tracker, recipe function, or browser parser for render_color_class was added.

Files: scripts/prepare_synthetic.py; scripts/multimodal_contract.py; scripts/train_spark_lora.py; tests/test_multimodal.py; README.md. Remote execution uses an isolated scripts/multimodal-v1 directory, preserving the original remote training script. [E1]

2. Dataset design and supervision

The supplied digital-twin corpus contains 600 PNGs: 12 render-control colors across 50 authored frame settings, all 640 by 480. Camera, lighting, and pose are fixed. Source image hashes and full decode were verified before staging. The corpus has 565 unique image hashes; animation frame is not measured volume. [E2]

Partition	Synthetic images	Reviewed real videos	Total
Training	427	21	448
Prepared validation	120	9	129
Actually evaluated	12	9	21

Synthetic splitting holds out the final ten authored frame settings (566-686), excludes two settings (539 and 552) as a gap, and deduplicates hashes within and across partitions. Fifty-three source records are excluded through the combined gap/deduplication policy. All 12 color controls occur in both partitions. Exact hash separation does not remove every visually similar scene.

Real replay retains the original 21 reviewed organizer clips; the same nine recipe-held-out clips remain regression examples. Their recording session is shared with training, so this remains development evaluation. No provisional field-label record was used. [E2, E3]

Target examples

Synthetic: **{"render_color_class": "navy"}**. The question explicitly asks for a synthetic appearance-control class from the 12-name palette. This is supervision from simulator metadata, not independent visual truth.

Real: **{"observed_color_sequence": ["blue", "dark green"]}**. Human-reviewed descriptions retain their original prompt and answer. No intended recipe or rendering-class name enters this real target.

Diagnostic selection and sampling

The bounded run evaluates all nine old real clips plus the first 12 synthetic validation records, one per color at frame 566. The other 108 synthetic validation images are prepared but not scored. --eval-limit 21 makes this explicit; omitting the limit evaluates the entire prepared set.

Twenty optimizer updates at accumulation four produce 80 presentations: 40 image and 40 video. This is not an epoch over 448 examples. The baseline, initial adapter, and continued adapter use identical selected inputs, prompts, preprocessing, and greedy 96-token generation. The explicit synthetic classification prompt differs from the earlier free-color diagnostic; their scores are not directly comparable.

3. Training readings and retained behavior

Reading	Measured value
Optimizer updates	20
Trainable parameters	7,667,712
Presentations	{"image": 40, "video": 40}
Total recorded job time	479.41 seconds (7.99 minutes), including loading, three evaluations and saves
Optimizer-loop elapsed sum	179.61 seconds; includes preprocessing/forward/backward
Peak allocated CUDA memory	19.54 GiB
First / last minibatch loss	0.5129 / 0.0324; different batches, not a fixed-set accuracy measure
Saved adapter SHA-256	792345ea39e4118c1f31907f38221ae3265b8cfb0c6480a12b5407d099d89a3e

These timings exclude dataset preparation, transfer, and preflight. They are not exclusive GPU-hours; other Spark workloads remained running. No energy or monetary cost was measured. Loss and gradient checks completed without non-finite values. [E3, E4]

Real-video regression cases

Vial	Target	Version 1	Continued adapter
04	green	green	green
05	orange	yellow > orange	orange
06	orange	orange	orange
13	blue > dark green	green	green > yellow
14	blue > dark green	green	green
15	blue > dark green	green	green
25	orange	orange	orange > brown
26	orange > brown	orange	orange > brown
27	orange	orange	orange > dark brown

4. Verification, risks, and next decisions

Verification completed

The local suite discovered 27 tests: 21 passed and six server tests were skipped because Flask is absent locally. New tests cover cross-split duplicate purging, class coverage, balanced replay, preservation of real regression cases, separate task metrics, and rejection of render-control labels on real observations. Spark preflight passed both image and video assistant-loss boundaries.

A real CUDA backward/optimizer run completed and produced a separate checkpoint. Saved tensors were checked for adapter-only names, finite values, shapes and parameter count; the source adapter hash remained unchanged. The original browser service was not restarted or replaced. [E1, E3, E4]

What the readings cannot establish

There is no untouched independent real-factory test in this experiment. Improved control-name agreement could reflect palette learning rather than better real appearance understanding. Continuing Version 1 with real replay also changes its real task exposure, so this experiment cannot isolate the benefit attributable to synthetic images. A matched real-only continuation and a mixed continuation are still required for that claim.

The synthetic scene omits lighting and camera diversity, realistic mixing/carryover targets, measured volume, and clear/water or brown control classes. The model may learn render-specific shortcuts or lose old shades and transitions despite replay. Exact shade-name agreement is stricter than broad visual similarity and is not a chemical metric.

Next release decision

Keep this adapter experimental. Inspect changes on the nine regression cases and score additional held-out synthetic states, then conduct a matched-budget real-only versus mixed-data comparison on independently verified real sessions. For production questions, add confirmed source/nozzle mapping, event labels, calibrated delivery measurements and backend recipe logic. Do not promote the adapter based solely on synthetic or same-session development scores.

Technical references

Hugging Face PEFT documents loading adapter weights with is_trainable=True: PEFT model API. This implementation deliberately starts a fresh optimizer.

Hugging Face documents native image/video inputs and grid tensors for the model architecture used by the extracted Cosmos reasoner: Qwen3-VL architecture API. These references describe APIs, not vial-task accuracy.

5. Reproduction and provenance

ID	Evidence
E1	Updated source, README.md, tests/test_multimodal.py, and the completed local test run.
E2	data/local/synthetic-mixed-v1/experiment.json, train.json, val.json; original synthetic archive manifest and original reviewed pilot. Private data excluded from Git.
E3	outputs/synthetic-mixed-v1/run.json, base.json, initial_adapter.json, adapted.json, and training.jsonl. Raw model predictions and run settings.
E4	outputs/synthetic-mixed-v1/artifact-verification.json and adapter-step-020; saved tensor checks and source-adapter integrity.

Command outline

Prepare with scripts/prepare_synthetic.py using --archive, --real-pilot and a fresh --out path. Use the Spark runner with --checkpoint base-reasoner --data data/synthetic-mixed-v1 --output runs/synthetic-mixed-v1 --initial-adapter runs/pilot-v1/adapter-step-020 --balance-modalities --eval-limit 21 --steps 20 --accumulation 4. Run the same command with --preflight before training. Full commands are in README.md.

Base and source identity

Base: nvidia/Cosmos3-Nano
Source snapshot: e59a53c25979a090fa8706c9acc0c254a6e89b92

Source Version 1 adapter SHA-256:
9dae56b981de7e7de200e39580e57cbcba81d3efdcf940f2e025e80a41a3a5f4

Synthetic source archive SHA-256:
67a3a065b169fdca053e8e86e0e5660a724db5ad9d9a9841a89a954a67ae088b

Trainer SHA-256:
c899dfdd2e9f5315158225022bde84a6667c680b8e860e5e991b4509a96ebe0f

Data/metric contract SHA-256:
015d97c325a284384740dc94b75976b069ef97d52567bb3155cfef9b6f462de9

Artifact locations and continuation

Spark experiment: /home/yt-vyflare71b7/workspace/vial-finetune-20261029/runs/synthetic-mixed-v1/. Local private copy: outputs/synthetic-mixed-v1/. The continued adapter is separate from pilot-v1/adapter-step-020. Always use a fresh output directory; never overwrite an evaluated checkpoint.

This is weight continuation with a fresh optimizer, not exact interrupted-run resume. Keep source identities and old checkpoints when incorporating later findings. The training path now supports genuine image supervision; its measured manufacturing value remains an empirical question.